Note in A General Family of Stochastic Proximal Gradient Methods for Deep Learning

Zhenwei Lin
2020213207
Research Institute for Interdisciplinary Science

Date: February 5, 2021

Abstract

This paper propose a unified framwork for stochastic proximal gradient descent, which they term PROX-GEN, that allows for arbitrary **positive preconditioners** and **lower semi-continuous regularizers**. They present two important update rules beyond the well-known standard methods as a byproduct of our approach: (i)the first closed-form proximal mappings of l_q regularization for adaptive stochastic gradient methods, (ii) a revised version of PROXQUANT that fixes a caveat of the original approach for quantization-specific regularizers.[1]

Keywords:

1 Introduction

We study the regularized training of neural networks, which can be formulated as the following(stochastic) optimization problem

$$\min_{\theta \in \omega} F(\theta) := \mathbb{E}_{\xi \sim \mathbb{P}}[f(\theta; \xi)] + \mathcal{R}(\theta) \tag{1}$$

where $\theta \in \mathbb{R}^p$ represents the network parameter vector, ξ is the random variable corresponding to mini-batch data samples, and $\mathcal{R}(\cdot)$ is a regularizer encouraging low-dimensional structural constraints on the parameter vector such as sparsity or low-rankness.

2 Background

3 Algorithm

Algorithm 1 PROXGEN: A General Stochastic Gradient Method

1: **procedure**

Input: Stepsize α_t,$\{\rho_t\}_{t=1}^{t=T} \in [0,1)$, regularization parameter λ, and small constant $0 < \delta \ll 1$

2: **Initialize:**$\theta_1 \in R^d$, $m_0 = 0$,and $C_0 = 0$

3: **for** t = 1,2,. . .,T **do**

4: Draw a minibatch sample ξ_t from

5: $g_t \leftarrow \nabla f(\theta_t;\xi_t)$

6: $m_t \leftarrow \rho_t m_{t-1} + (1-\rho_t) g_t$

7: $C_t \leftarrow$Preconditioner construction

8: $\theta_{t+1} \in argmin_{\theta\in\Omega}\left\{\langle m_t,\theta\rangle + \lambda R(\theta) + \frac{1}{2\alpha_t}(\theta-\theta_t)^T(C_t+\delta I)(\theta-\theta_t)\right\}$

9: **Output:**θ_{T+1}

The update rule on line8 of 1 can be written more compactly:

$$\begin{aligned}\theta_{t+1} &\in argmin_{\theta\in\Omega}\left\{\langle m_t,\theta\rangle + \lambda R(\theta) + \frac{1}{2\alpha_t}(\theta-\theta_t)^T(C_t+\delta I)(\theta-\theta_t)\right\} \\ &= prox_{\alpha_t\lambda R(\cdot)}^{C_t+\delta I}\left(\theta_t - \alpha_t(C_t+\delta I)^{-1}m_t\right)\end{aligned} \tag{2}$$

where the proximal operator in 2 is defined as $prox_h^A(z) = argmin_x\left\{h(x) + \frac{1}{2}\|x-z\|_A^2\right\}$.

Then it follows proximal operator that 2's second term can be written as follows.

$$\begin{aligned}&prox_{\alpha_t\lambda R(\cdot)}^{C_t+\delta I}\left(\theta_t - \alpha_t(C_t+\delta I)^{-1}m_t\right) \\ =&argmin_\theta\left\{\alpha_t\lambda R(\theta) + \frac{1}{2}\|\theta-(\theta_t-\alpha_t(C_t+\delta I)^{-1}m_t)\|_{C_t+\delta I}^2\right\} \\ =&argmin_\theta\left\{\alpha_t\lambda R(\theta) + \frac{1}{2}\left\langle\theta-(\theta_t-\alpha_t(C_t+\delta I)^{-1}m_t),(C_t+\delta I)(\theta-(\theta_t-\alpha_t(C_t+\delta I)^{-1}m_t))\right\rangle\right\} \\ =&argmin_\theta\left\{\alpha_t\lambda R(\theta) + \frac{1}{2}\left\langle(\theta-\theta_t)+\alpha_t(C_t+\delta I)^{-1}m_t,(C_t+\delta I)(\theta-\theta_t)+\alpha_t m_t\right\rangle\right\} \\ =&argmin_\theta\left\{\alpha_t\lambda R(\theta) + \frac{1}{2}(\theta-\theta_t)^T(c_t+\delta I)(\theta-\theta_t) + 2(\theta-\theta_t)^T(\alpha_t m_t) + constant\right\} \\ =&argmin_\theta\left\{\lambda R(\theta) + \frac{1}{2\alpha_t}(\theta-\theta_t)^T(c_t+\delta I)(\theta-\theta_t) + 2(\theta-\theta_t)^T(m_t) + constant\right\}\end{aligned} \tag{3}$$

Relationship with PROX-SGD

PROX-SGD is not an exact proximal approach and is significantly different from our **PROXGEN**. Unlike **PROGEN** updating parameters by directly solving the quadratic sub-problem. **PROX-SGD** updates the parameters in two stage: solving the sub-problem and then updating the parameters with the computed

direction by learning rate, i.e.,

$$\begin{aligned} \hat{\theta}_t &= \underset{\theta \in \Omega}{argmin} \left\{ \langle m_t, \theta \rangle + \lambda R(\theta) + \frac{1}{2}(\theta - \theta_t)^T (C_t + \delta I)(\theta - \theta_t) \right\} \\ \theta_{t+1} &= \theta_t + \alpha_t(\hat{\theta}_t - \theta_t) \end{aligned} \tag{4}$$

Remark: The two-stage update scheme of **PROX-SGD** might have some potential issues. For example.

1. for l_1-regularized problems, the upadte parameter θ_{t+1} in 4 might not achieve exact zero (while $\hat{\theta}_t$ can) whereas θ_{t+1} for **PROGEN** can attain exact zero value according to the update rule
2. **PROX-SGD** might overestimate the sparsity level.

4 Convergence Analysis

Definition 4.1 (Fréchet Subdifferential)**.** Let φ be a real-valued function. The Fréchet Subdifferential of φ at $\bar{\theta}$ with $|\varphi(\bar{\theta})| < \infty$ is defined by

$$\widehat{\partial}\varphi(\bar{x}) := \left\{ \theta^* \in \Omega \mid \liminf_{\theta \to \bar{\theta}} \frac{\varphi(\theta) - \varphi(\bar{\theta}) - \left\langle \theta^*, \theta - \bar{\theta} \right\rangle}{\|\theta - \bar{\theta}\|} \geq 0 \right\} \tag{5}$$

To derive the convergence bound, we make the following mild assumptions:

1. (L-smoothness) The loss function f is differentiable, $L-$smooth, and lower-bounded:
$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \tag{6}$$
2. (Bounded variance) The stochastic gradient $g_t = \nabla f(\theta; \xi)$ is unbiased and has the bounded variance:
$$\mathbb{E}_\xi[\nabla f(\theta_t; \xi)] = \nabla f(\theta_t), \mathbb{E}_\xi[\|g_t - \nabla f(\theta_t)\|^2] \leq \sigma^2 \tag{7}$$
3. (i) final step-vector is finite. (ii) the stochastic gradient is bounded, and (iii) the momentum parameter should be exponentially decaying:
$$\|\theta_{t+1} - \theta_t\| \leq D, \quad \|g_t\| \leq G, \quad \rho_t = \rho_0 \mu^{t-1} \tag{8}$$
with $D, G > 0$ and $\rho_0, \mu \in [0, 1)$
4. (Sufficiently positive-definite) The minimum eigenvalue of effective spectrums should be uniformly lower bounded over all time t by some strictly positive constant γ:
$$\lambda_{min}(\alpha_t(C_t + \delta I)^{-1}) \geq \gamma > 0 \tag{9}$$

Conditions 6 and 7 are standard in general non-convex optimization. In addition, condition8 is extensively studied in previous literature in the context of adaptive gradient methods. Lastly, a similar condition to 9 is also considered and it can be easily satisfied in practice.

Here, we use the following covergence criterion $\mathbb{E}[dist(0, \hat{\partial}F(\theta))] \leq \epsilon$ for an $\epsilon-$ stationary point. If no regularizer is considered ($\mathcal{R} = 0$), this criterion boils down to the one usually used in non-convex optimization, $\mathbb{E}[\|\nabla f(\theta)\|] \leq \epsilon$.

We are now ready to state our main theorem for general convergence.

Theorem 4.1. *Let θ_a denote an iterate uniformly randomly chosen from $[\theta_1, \cdots, \theta_T]$. Under the condition 6,7,8,9, with the initial stepsize $\alpha_0 \leq \frac{\delta}{3L}$ and non-increasing stepsize α_t,PROXGEN, algorithm1 is guaranteed to yield*

$$\mathbb{E}[dist(0, \hat{\partial} F(\theta_a))^2] \leq \frac{Q_1 \sigma^2}{T} \sum_{t=0}^{T-1} \frac{1}{b_t} + \frac{Q_2 \Delta}{T} + \frac{Q_3}{T} \tag{10}$$

where $\Delta = f(\theta) - f(\theta^*)$ with optimal point θ^*, and b_t ***is the minibatch size*** *at time t. The constanats $\{Q_i\}_{i=1}^3$ on the right-hand side depend on the constants $\{\alpha_0, L, D, G, \rho_0, \mu, \gamma\}$ but not on T.*

Intuition: It can be seen that the appropriate minibatch size is important to ensure a good convergence rate. Various setting for the minibatch size could be employed for convergence guarantee(for example, dynamic minibatch size $b_t = t$), but in order to consider practical cases, we provide the following important corollary for constant minibatch size.

Corollary (Constant Mini-batch)**.** *Under the same assumptions as in Theorem 4.1 with constant minibatch size $b_t = b = \Theta(T)$, we have $\mathbb{E}[dist(0, \hat{\partial} F(\theta_a))^2] \leq O(1/T)$ and the total complexity is $O(1/\epsilon^4)$ in order to have $\mathbb{E}[dist(0, \hat{\partial} F(\theta_a))] \leq \epsilon$*

Remarks:

******

Lemma 4.2. *The first-order momentum m_t generated by $m_t = (1 - \rho_t) m_{t-1} + \rho_t g_t$ in Algorithm1 has an upper bound, i.e.,*

$$\|m_t\| \leq G \tag{11}$$

where G is from the inequality $\|g\| \leq G$

Proof. By mathematical induction. For $t = 1$, the momentum is computed as $m_1 = \rho_1 m_0 + (1 - \rho_1) g_1 = (1 - \rho_0) g_1$. Therefore, we have

$$\|m_1\|_2 = \|(1 - \rho_0) g_1\| \leq (1 - \rho_0) G \leq G. \tag{12}$$

Now, we assume that $\|m_{t-1}\|_2 \leq G$ holds. The momentum at time t is constructed by $m_t = (1 - \rho_t) m_{t-1} + \rho_t g_t$

$$\begin{aligned} \|m_t\|_2 &= \|(1 - \rho_t) m_{t-1} + \rho_t g_t\|_2 \\ &\leq (1 - \rho_t) \|m_{t-1}\|_2 + \rho_t \|g_t\|_2 \\ &\leq (1 - \rho_t) G + \rho_t G = G \end{aligned} \tag{13}$$

where the first inequality comes from the triangle inequality and the second one is derived from the induction hypothesis. □

4.1 Proof of Main Theorem

Step 1: 1. First, we have to clarify our goal is to prove the distance between 0 and subgradient is small enough, i.e., $dist(0, \hat{\partial} F(\theta_{t+1}))^2$. So the first step is to modify the goal $\hat{\partial} F$.

From the algorithm1, we know the sequence generated by a minimize an quadratic approximation function

$$\theta_{t+1} \in \underset{\theta}{argmin} \left\{ \langle (1-\rho_t) g_t + \rho_t m_{t-1}, \theta \rangle + \mathcal{R}(\theta) + \frac{1}{2\alpha_t} (\theta - \theta_t)^T (C_t + \delta I)(\theta - \theta_t) \right\} \tag{14}$$

By the approximation function's optimality condition, we have

$$0 \in (1-\rho_t) g_t + \rho_t m_{t-1} + \hat{\partial} \mathcal{R}(\theta_{t+1}) + \frac{1}{\alpha_t} (C_t + \delta I)(\theta_{t+1} - \theta_t) \tag{15}$$

which means that

$$-(1-\rho_t) g_t - \rho_t m_{t-1} - \frac{1}{\alpha_t} (C_t + \delta I)(\theta_{t+1} - \theta_t) \in \hat{\partial} \mathcal{R}(\theta_{t+1}) \tag{16}$$

By adding the gradient $\nabla f(\theta_{t+1})$ on the both side, we have

$$\nabla f(\theta_{t+1}) - (1-\rho_t) g_t - \rho_t m_{t-1} - \frac{1}{\alpha_t} (C_t + \delta I)(\theta_{t+1} - \theta_t) \in \nabla f(\theta_{t+1}) + \hat{\partial} \mathcal{R}(\theta_{t+1}) = \hat{\partial} F(\theta_{t+1}) \tag{17}$$

□

Step 2:

$$\begin{aligned}
& dist(0, \hat{\partial} F(\theta_{t+1}))^2 \\
=& \|(1-\rho_t) g_t - \nabla f(\theta_{t+1}) + \rho_t m_{t-1} + \frac{1}{\alpha_t} (C_t + \delta I)(\theta_{t+1} - \theta_t)\|_2^2 \\
=& \|(1-\rho_t) g_t - \nabla f(\theta_{t+1}) + \rho_t m_{t-1} + (\theta_{t+1} - \theta_t) + \frac{1}{\alpha_t} (C_t + \delta I)(\theta_{t+1} - \theta_t) - (\theta_{t+1} - \theta_t)\|_2^2 \\
\leq& 3\|(1-\rho_t) g_t - \nabla f(\theta_{t+1}) + \rho_t m_{t-1} + (\theta_{t+1} - \theta_t)\|_2^2 + 3\|\frac{1}{\alpha_t} (C_t + \delta I)(\theta_{t+1} - \theta_t)\|_2^2 \\
& + 3\|(\theta_{t+1} - \theta_t)\|_2^2 \\
\leq& 3\|(1-\rho_t) g_t - \nabla f(\theta_{t+1}) + \rho_t m_{t-1} + (\theta_{t+1} - \theta_t)\|_2^2 + 3(\frac{1}{\gamma^2} + 1)\|\theta_{t+1} - \theta_t\|_2^2
\end{aligned} \tag{18}$$

The first equality is from step1, and the second equality is by addition and subtraction. The first inequality is by $(a+b+c)^2 \leq 3a^2 + 3b^2 + 3c^2$, and the second inequality is by an assumptions $\lambda_{max}(\frac{1}{\alpha_t}(C_t + \delta I)) \leq \frac{1}{\gamma}$ which yields from the assumption 9.

Now we denote $\|(1-\rho_t) g_t - \nabla f(\theta_{t+1} + \rho_t m_{t-1})\|_2^2 + \|\theta_{t+1} - \theta_t\|_2^2$ as T_1

Then we have

$$\begin{aligned}
T_1 &= \|(1-\rho_t) g_t - \nabla f(\theta_{t+1}) + \rho_t m_{t-1} + \theta_{t+1} - \theta_t\|_2^2 \\
&= \|(1-\rho_t) g_t - \nabla f(\theta_{t+1}) + \rho_t m_{t-1}\|_2^2 + \|\theta_{t+1} - \theta_t\|_2^2 \\
&\quad + 2 \langle (1-\rho_t) g_t - \nabla f(\theta_{t+1}) + \rho_t m_{t-1}, \theta_{t+1} - \theta_t \rangle
\end{aligned} \tag{19}$$

Next we need to consider how to find an upper bound to bound the third term.

□

Step 3: Denote $\langle (1-\rho_t)g_t + \rho_t m_{t-1}, \theta\rangle + \mathcal{R}(\theta) + \frac{1}{2\alpha_t}(\theta-\theta_t)^T(C_t+\delta I)(\theta-\theta_t)$ as $M_t(\theta)$ Using θ_{t+1} is the minimum point in step t, then by $M_t(\theta_{t+1}) \le M_t(\theta_t)$, we obtain

$$\begin{aligned}&\langle (1-\rho_t)g_t + \rho_t m_{t-1}, \theta_{t+1}\rangle + \mathcal{R}(\theta_{t+1}) + \frac{1}{2\alpha_t}(\theta_{t+1}-\theta_t)^T(C_t+\delta I)(\theta_{t+1}-\theta_t)\\ \le &\langle (1-\rho_t)g_t + \rho_t m_{t-1}, \theta_t\rangle + \mathcal{R}(\theta_t)\end{aligned} \tag{20}$$

which in result

$$\langle (1-\rho_t)g_t + \rho_t m_{t-1}, \theta_{t+1}-\theta_t\rangle + \mathcal{R}(\theta_{t+1}) + \frac{1}{2\alpha_t}(\theta_{t+1}-\theta_t)^T(C_t+\delta I)(\theta_{t+1}-\theta_t) \le \mathcal{R}(\theta_t) \tag{21}$$

By decent lemma, we have

$$f(\theta_{t+1}) \le f(\theta_t) + \langle \nabla f(\theta_t), \theta_{t+1}-\theta_t\rangle + \frac{L}{2}\|\theta_{t+1}-\theta_t\|_2^2 \tag{22}$$

Adding previous two inequalities yields

$$\langle (1-\rho_t)g_t - \nabla f(\theta_t) + \rho_t m_{t-1}, \theta_{t+1}-\theta_t\rangle + (\theta_{t+1}-\theta_t)^T\left(\frac{1}{2\alpha_t}(C_t+\delta I) - \frac{L}{2}I\right)(\theta_{t+1}-\theta_t) \le F(\theta_t) - F(\theta_{t+1}) \tag{23}$$

Therefore, we have

$$\begin{aligned}&\langle (1-\rho_t)g_t - \nabla f(\theta_{t+1}) + \rho_t m_{t-1}, \theta_{t+1}-\theta_t\rangle\\ \le &F(\theta_t) - F(\theta_{t+1}) - \langle \nabla f(\theta_{t+1}) - \nabla f(\theta_t), \theta_{t+1}-\theta_t\rangle - \|\theta_{t+1}-\theta_t\|^2_{\frac{1}{2\alpha_t}(C_t+\delta I)-\frac{L}{2}I}\\ \le &F(\theta_t) - F(\theta_{t+1}) - \langle \nabla f(\theta_{t+1}) - \nabla f(\theta_t), \theta_{t+1}-\theta_t\rangle + \left(\frac{\delta}{2\alpha_0} - \frac{3L}{2}\right)\|\theta_{t+1}-\theta_t\|_2^2\end{aligned} \tag{24}$$

The second inequality is by $\frac{1}{2\alpha_t}(C_t+\delta I) - \frac{3}{2}LI \ge \frac{1}{2\alpha_0}\delta I - \frac{3}{2}LI$, since C_t is positive(semi) definite and α_t is non-increasing(learning rate decaying). Therefore, from this we can derive the stepsize condition in our Theorem as $\alpha_0 \le \frac{\delta}{3L}$, then we have

$$\frac{1}{2\alpha_t}(C_t+\delta I) - \frac{3}{2}LI \ge \frac{1}{2\alpha_0}\delta I - \frac{3}{2}LI \ge 0 \tag{25}$$

Since 25, the second inequality is reasonable.

Then we have some inequalities about T_1.

$$\begin{aligned}T_1 =&\|(1-\rho_t)g_t - \nabla f(\theta_{t+1}) + \rho_t m_{t-1}\|_2^2 + \|\theta_{t+1}-\theta_t\|_2^2\\ &+ 2\langle (1-\rho_t)g_t - \nabla f(\theta_{t+1}) + \rho_t m_{t-1}, \theta_{t+1}-\theta_t\rangle\\ \le&\|(1-\rho_t)g_t - \nabla f(\theta_{t+1}) + \rho_t m_{t-1}\|_2^2 + \|\theta_{t+1}-\theta_t\|_2^2\\ &+ F(\theta_t) - F(\theta_{t+1}) - \langle \nabla f(\theta_{t+1}) - \nabla f(\theta_t), \theta_{t+1}-\theta_t\rangle + \left(\frac{\delta}{2\alpha_0} - \frac{3L}{2}\right)\|\theta_{t+1}-\theta_t\|^2\\ \le&4\|g_t - \nabla f(\theta_t)\|_2^2 + 4L^2\|\theta_{t+1}-\theta_t\|_2^2 + 4\|\rho_t m_{t-1}\|_2^2 + 4\|\rho_t g_t\|_2^2 + \|\theta_{t+1}-\theta_t\|^2\\ &+ F(\theta_t) - F(\theta_{t+1}) + L\|\theta_{t+1}-\theta_t\|_2^2 + \left(\frac{\delta}{2\alpha_0} - \frac{3L}{2}\right)\|\theta_{t+1}-\theta_t\|^2\\ \le&F(\theta_t) - F(\theta_{t+1}) + 4\rho_0^2\mu^{2(t-1)}G^2 + 4\rho_0^2\mu^{2(t-1)}G^2\\ &+ \left(\frac{\delta}{2\alpha_0} + 1 + 4L^2 - \frac{L}{2}\right)\|\theta_{t+1}-\theta_t\|_2^2 + 4\|g_t - \nabla f(\theta_t)\|\end{aligned} \tag{26}$$

Now let me return to 18, we have

$$
\begin{aligned}
&dist(0, \hat{\partial} F(\theta_{t+1}))^2 \\
\leq & 3\left(F(\theta_t) - F(\theta_{t+1}) + 8\rho_0^2\mu^{2(t-1)}G^2 + \left(\frac{\delta}{2\alpha_0} - \frac{L}{2} + 2 + 4L^2 + \frac{1}{\gamma^2}\right)\|\theta_{t+1} - \theta_t\|_2^2 + 4\|g_t - \nabla f(\theta_t)\|_2^2\right)
\end{aligned} \tag{27}
$$

Here we denote $\left(\frac{\delta}{2\alpha_0} - \frac{L}{2} + 2 + 4L^2 + \frac{1}{\gamma^2}\right)$,$f(\theta_0) - f(\theta_*)$ as C_2 and Δ .

Then we have

$$
\begin{aligned}
&\mathbb{E}[dist(0, \hat{\partial} F(\theta_a))^2] \\
\leq & \frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\|(1-\rho_t)g_t - \nabla f(\theta_{t+1}) + \rho_t m_{t-1} + \frac{1}{\alpha_t}(C_t + \delta I)(\theta_{t+1} - \theta_t)\|_2^2\right] \\
\leq & \frac{3}{T}\left(\Delta + \frac{8\rho_0^2 G^2}{1-\mu^2} + 4\sum_{t=0}^{T-1}\|g_t - \nabla f(\theta_t)\|_2^2 + C_2\sum_{t=0}^{T-1}\|\theta_{t+1} - \theta_t\|_2^2\right)
\end{aligned} \tag{28}
$$

So the next step is to find an upper bound to bound $\sum_{t=0}^{T-1}\|\theta_{t+1} - \theta_t\|_2^2$

□

Step 4: The upper bound from the inequality

$$
\left(\frac{\delta}{2\alpha_0} - \frac{3}{2}L\right)\sum_{t=0}^{T-1}\|\theta_{t+1} - \theta_t\|_2^2 \leq \sum_{t=0}^{T-1}\|\theta_{t+1} - \theta_t\|^2_{\frac{1}{2\alpha_t}(C_t+\delta I) - \frac{3}{2}LI} \tag{29}
$$

From 23 we have

$$
\begin{aligned}
&\|\theta_{t+1} - \theta_t\|^2_{\frac{1}{2\alpha_t}(C_t+\delta I) - \frac{3}{2}LI} \\
\leq & F(\theta_t) - F(\theta_{t+1}) - \langle (1-\rho_t)g_t - \nabla f(\theta_t), \theta_{t+1} - \theta_t\rangle - \langle \rho_t m_{t-1}, \theta_{t+1} - \theta_t\rangle \\
= & F(\theta_t) - F(\theta_{t+1}) - \langle g_t - \nabla f(\theta_t), \theta_{t+1} - \theta_t\rangle + \langle \rho_t g_t, \theta_{t+1} - \theta_t\rangle - \langle \rho_t m_{t-1}, \theta_{t+1} - \theta_t\rangle \\
= & F(\theta_t) - F(\theta_{t+1}) - \left\langle \frac{1}{\sqrt{L}}(g_t - \nabla f(\theta_t)), \sqrt{L}(\theta_{t+1} - \theta_t)\right\rangle + \left\langle \frac{1}{\sqrt{L}}(\rho_t g_t), \sqrt{L}\theta_{t+1} - \theta_t\right\rangle - \langle \rho_t m_{t-1}, \theta_{t+1} - \theta_t\rangle \\
\leq & F(\theta_t) - F(\theta_{t+1}) + \frac{1}{2L}\|g_t - \nabla f(\theta_t)\|_2^2 + \frac{L}{2}\|\theta_{t+1} - \theta_t\|_2^2 + \frac{\rho_t^2}{2L}\|g_t\|^2 + \frac{L}{2}\|\theta_{t+1} - \theta_t\|_2^2 + \|\rho_t m_{t-1}\|_2\|\theta_{t+1} - \theta_t\|_2 \\
\leq & F(\theta_t) - F(\theta_{t+1}) + \rho_0\mu^{t-1}DG + \frac{\rho_0^2\mu^{2(t-1)}G^2}{2L} + L\|\theta_{t+1} - \theta_t\|_2^2 + \frac{1}{2L}\|g_t - \nabla f(\theta_t)\|_2^2
\end{aligned} \tag{30}
$$

The deriviations in inequalities as follows:

1. We rearrange the inequality.
2. we use the fact $\langle a, b\rangle = \left\langle ca, \frac{1}{c}b\right\rangle \leq c^2\|a\|_2^2 + \frac{1}{c^2}\|b\|_2^2$ for any positive constant c. and Cauchy-Schwartz inequality $\langle a, b\rangle \leq \|a\|_2\|b\|_2$
3. We apply our assumption and Lemma 4.2

Therefore, we have

$$
\begin{aligned}
&\sum_{t=0}^{T-1} \|\theta_{t+1} - \theta_t\|^2_{\frac{1}{2\alpha}(C_t+\delta I)-\frac{3}{2}LI} \\
\leq & F(\theta_0) - F(\theta_*) + \frac{\rho_0 DG}{1-\mu} + \frac{\rho_0^2 G^2}{2L(1-\mu^2)} + \frac{1}{2L}\sum_{t=0}^{T-1} \|g_t - \nabla f((\theta_t))\|_2^2 \\
= & \Delta + C_1 + \frac{1}{2L}\sum_{t=0}^{T-1} \|g_t - \nabla f((\theta_t))\|_2^2
\end{aligned}
\tag{31}
$$

Futhermore, by 29, we obtain

$$
\sum_{t=0}^{T-1} \|\theta_{t+1} - \theta_t\|_2^2 \leq H_1 + H_2 \sum_{t=0}^{T-1} \|g_t - \nabla f(\theta_t)\|_2^2 \tag{32}
$$

where the constants H_1 and H_2 are defined as

$$
\begin{aligned}
H_1 = & \frac{\Delta}{\left(\frac{\delta}{2\alpha_0} - \frac{3}{2}L\right)} + \frac{C_1}{\left(\frac{\delta}{2\alpha_0} - \frac{3}{2}L\right)} \\
H_2 = & \frac{1}{2L\left(\frac{\delta}{2\alpha_0} - \frac{3}{2}L\right)}
\end{aligned}
\tag{33}
$$

Now let us go back to formula 28

$$
\begin{aligned}
&\mathbb{E}[dist(0, \hat{\partial} F(\theta_a))^2] \\
\leq & \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\left[\|(1-\rho_t)g_t - \nabla f(\theta_{t+1}) + \rho_t m_{t-1} + \frac{1}{\alpha_t}(C_t + \delta I)(\theta_{t+1} - \theta_t)\|_2^2\right] \\
\leq & \frac{3}{T}\left(\Delta + \frac{8\rho_0^2 G^2}{1-\mu^2} + 4\sum_{t=0}^{T-1} \|g_t - \nabla f(\theta_t)\|_2^2 + C_2 \sum_{t=0}^{T-1} \|\theta_{t+1} - \theta_t\|_2^2\right) \\
\leq & \frac{3}{T}\left(\Delta + \frac{8\rho_0^2 G^2}{1-\mu^2} + 4\sum_{t=0}^{T-1} \|g_t - \nabla f(\theta_t)\|_2^2 + C_2\left(H_1 + H_2 \sum_{t=0}^{T-1} \|g_t - \nabla f(\theta_t)\|_2^2\right)\right) \\
\leq & \frac{Q_1}{T}\sum_{t=0}^{T-1} \mathbb{E}\left[\|g_t - \nabla f(\theta_t)\|_2^2\right] + \frac{Q_2 \Delta}{T} + \frac{Q_3}{T}
\end{aligned}
\tag{34}
$$

where

$$
Q_1 = 4 + C_2 H_2, \quad Q_2 = 3 + \frac{3C_2}{\frac{\delta}{2\alpha_0} - \frac{3}{2}L}, \quad Q_3 = \frac{24\rho_0^2 G^2}{1-\mu^2} + \frac{3C_1 C_2}{\frac{\delta}{2\alpha_0} - \frac{3}{2}L} \tag{35}
$$

Then only the last item left $\mathbb{E}\left[\|g_t - \nabla f(\theta_t)\|_2^2\right]$ which could be thought as mean's variance is needed to

deal with.

$$
\begin{aligned}
\|g_t - \nabla f(\theta_t)\|_2^2 &= \mathbb{E}\left[\| \left(\frac{1}{b_t}\sum_{i=1}^{b_t} \nabla f\left(\theta_t;\xi_{i_t}\right)\right) - \nabla f(\theta_t)\right] \|_2^2\Bigg] \\
&= \frac{1}{b_t^2}\mathbb{E}\left[\left\|\sum_{i=1}^{b_t}\left\{\nabla f\left(\theta_t;\xi_{i_t}\right) - \nabla f(\theta_t)\right\}\right\|_2^2\right] \\
&\leq \frac{1}{b_t^2}\sum_{i=1}^{b_t}\mathbb{E}\left[\left\|\nabla f\left(\theta_t;\xi_{i_t}\right) - \nabla f(\theta_t)\right\|_2^2\right] \\
&\leq \frac{1}{b_t}\sigma^2
\end{aligned}
\tag{36}
$$

So far, we complete our proof.

$$
\mathbb{E}_R\left[\operatorname{dist}\left(\mathbf{0}, \widehat{\partial} F(\theta_R)\right)^2\right] \leq \frac{Q_1\sigma^2}{T}\sum_{t=0}^{T-1}\frac{1}{b_t} + \frac{Q_2\Delta}{T} + \frac{Q_3}{T} \tag{37}
$$

□

References

[1] J. Yun, A. C. Lozano, and E. Yang, *A general family of stochastic proximal gradient methods for deep learning*, arXiv preprint arXiv:2007.07484, (2020).